

Abstract

Motivation

Paper-to-poster systems can generate strong first drafts, but refinement remains difficult.

Challenge

Users often know a poster feels unclear or imbalanced, but cannot precisely specify what is wrong or how to fix it.

Idea

PROS reframes poster improvement as proactive multimodal refinement rather than reactive instruction following.

Contribution

Given a poster draft and its source paper, PROS diagnoses repairable deficiencies, assigns them to executable refinement stages, and applies grounded edits over editable poster states.

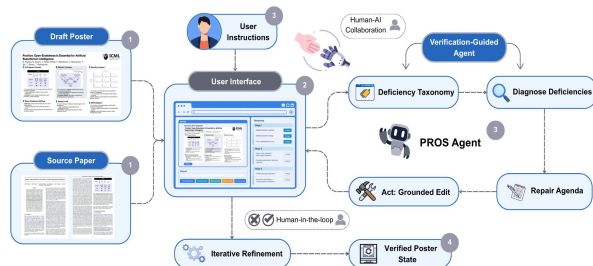
Takeaway

PROS helps users discover, understand, and repair poster issues through a controllable human-AI editing loop.

Key Insight

- PROS reframes poster refinement as a proactive, structured optimization problem.
- Instead of only executing user instructions, PROS helps identify, prioritize, and repair deficiencies over editable poster states.

Methods



PROS has three components

1. Grounded editing substrate — Editable PPTX objects are aligned with semantic blocks and rendered visual regions.
2. Executable taxonomy — Six repairable deficiency classes cover structure, layout, and content.
3. Stagewise refinement — PROS fixes structural failures first, then rebalances layout, then improves paper-grounded communication.

Structure → Layout → Content

Three-Stage Refinement Summary

Stage	Taxonomy Class
Stage 1	Overflow / out-of-bounds; overlap / collision
Stage 2	Excessive whitespace / blank-region imbalance; section allocation / layout imbalance
Stage 3	Content insufficiency; content redundancy / noise / overload

Results

Repair Performance

- PROS resolves 73.0% of active deficiencies on the 16-case editable subset.
- Repair performance is strongest for structural validity failures.
- Stagewise refinement reduces layout and content-level issues without breaking editable poster structure.

Holistic Evaluation

- Average VLM-judged poster quality improves from 3.1 to 4.0.
- Average VLM-judged holistic gain reaches +29.1% across 8 matched before/after cases.
- Improvements are visible in spacing, section balance, and paper-grounded communication.

Diagnosis Performance

- PROS achieves 82.0% macro diagnosis F1 across the six executable taxonomy classes.
- Diagnosis is strongest for whitespace and section-allocation imbalance.

Table 1. Preliminary repair performance on the current 16-case mixed subset (8 PosterGen + 8 Paper2Poster)

Stage	Taxonomy	Cases before	Cases after	Resolution
Stage 1	Overflow / out-of-bounds	8	0	1.00
	Overlap / collision	18	4	0.78
	Whitepace / blank-region imbalance	14	6	0.57
Stage 2	Section allocation / layout imbalance	11	1	0.91
	Content insufficiency	16	7	0.56
Stage 3	Content redundancy / noise / overload	12	5	0.58
Average repair resolution		—	—	0.73

Table 2. Pilot VLM-as-a-judge evaluation on all 8 matched before/after cases in the current editable subset

Source	#Cases	Overall before	Overall After	Overall Gain	Gain (%)
PosterGen	4	3.3	4.1	+0.8	23.4%
Paper2Poster	4	2.8	3.9	+1.1	36.1%
Overall aver-	8	3.1	4.0	+0.9	29.1%

Discussion

Why proactive refinement matters

- Reactive editing assumes users already know the right instruction. In poster refinement, users often only know that the draft is unclear, crowded, or incomplete.
- PROS addresses this gap by helping users identify what should be improved, why it matters, and how it can be repaired.
- This makes poster editing a collaborative diagnosis-and-repair process instead of a one-shot generation task.

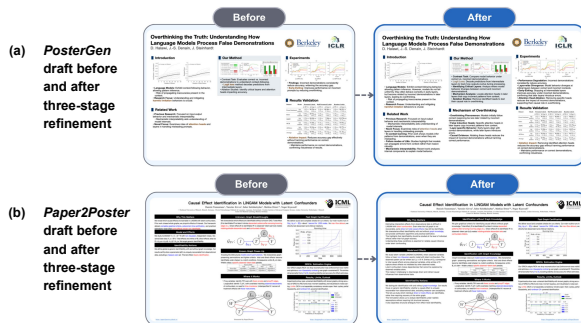
Conclusions

- PROS introduces proactive refinement for scientific posters.
- It combines multimodal diagnosis, executable taxonomy, and staged grounded repair.
- Preliminary results show 82.0% diagnosis F1, 73.0% repair resolution, and +29.1% VLM-judged holistic quality gain.
- Proactive refinement supports controllable human-AI editing of structured visual artifacts.

References

- Pang et al., *Paper2Poster*, 2025.
- Zhang et al., *PosterGen*, 2025.
- Shi et al., *APEX*, 2026.

Representative Before/After Refinement Examples



Contact

Xingda Lyu
University of Washington
Email: xlyu18@uw.edu